

- Theme : MedTech / BioTech / HealthTech



- **Problem Statement Title:** Efficient systems for early detection, prevention, and management of livestock diseases and animal health issues..
- **Problem Statement ID:** SIH26128
- **PS Category:** Software (Web + AI Bot)
 - Team ID -
 - Team Name : Neural Knights

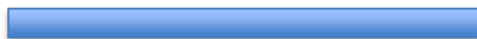


Pashusahay,, Empowering Livestock Care with AI & Instant Emergency Assistance



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- Our project, Pashusahay, is a Web-based Livestock Surveillance Platform, driven by a WhatsApp AI Agent. “People can just report animal health problems through WhatsApp in local languages to get immediate advice. This data is automatically fed into our Web Dashboard to track disease clusters and inform veterinarians and government officials in real time to avoid epidemics



- **Core Platform:** ,PashuSahay is a web-based livestock surveillance platform driven by a WhatsApp AI agent.
- **Data Collection & Guidance:** Anyone can report animal health issues on WhatsApp in their local language and get fast live AI guidance.
- **Web Processing & Outbreak Prevention:** The data collected goes directly into our Web Dashboard to track illness clusters and warn veterinarians and government officials in real time..

Technology Stack

Frontend

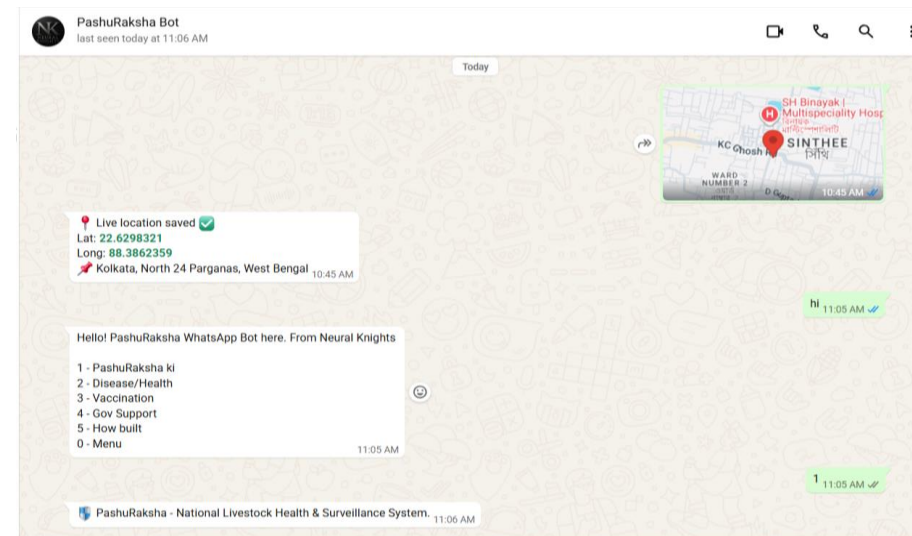
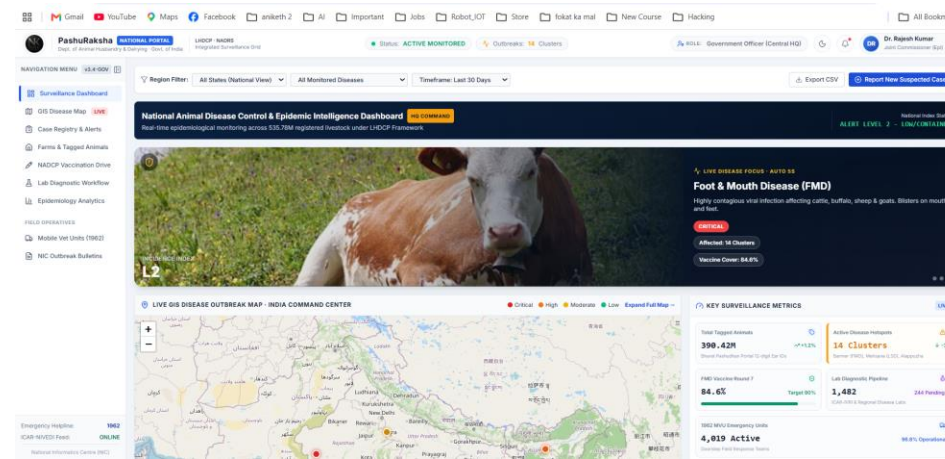
React
TypeScript
Tailwind Css

Backend

Java
PostgreSql
Python
Node js

Developer Tools

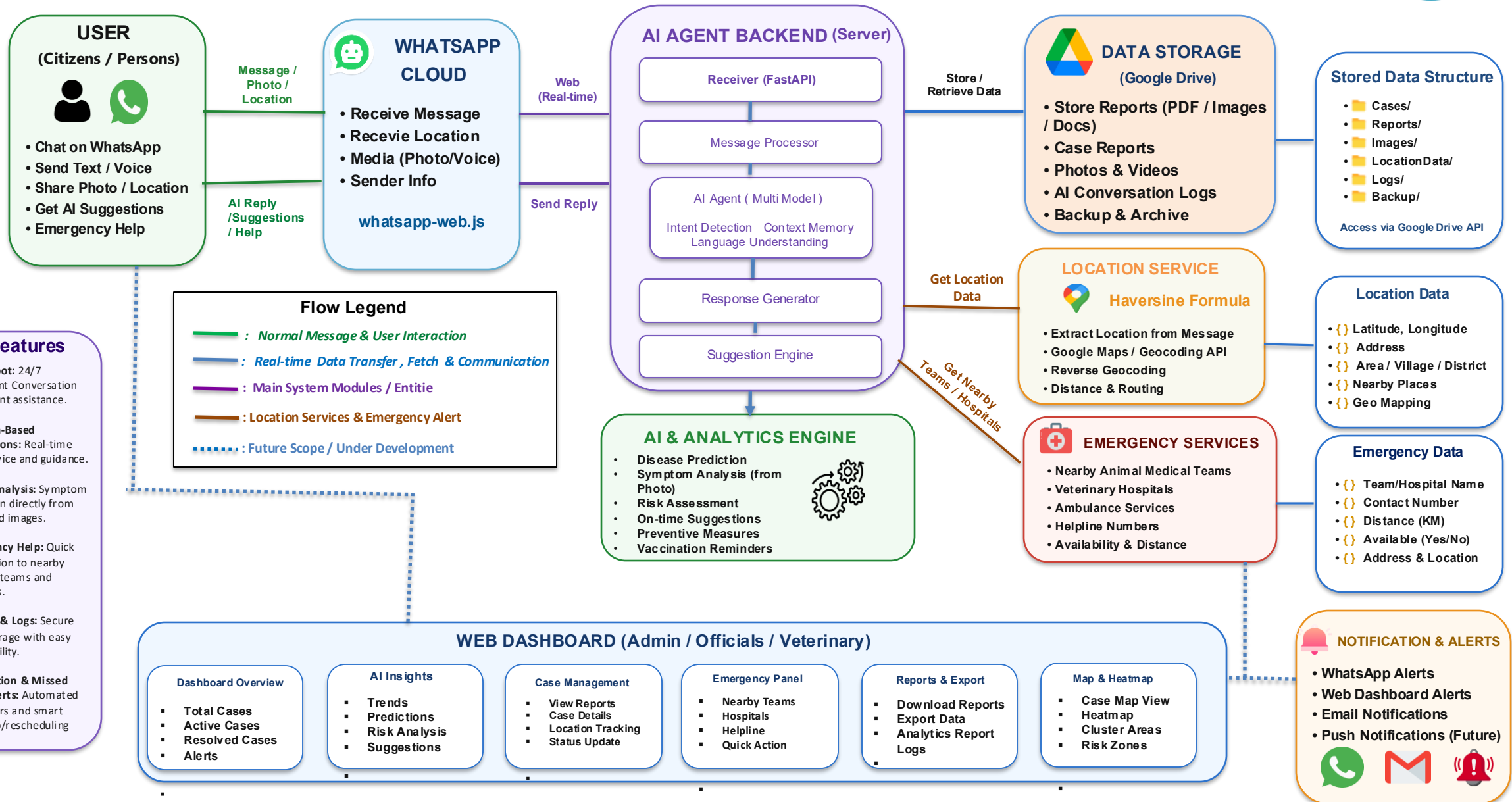
Vs Code
Git / Github
Docker



System Architecture & Workflow



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Practicality & Growth Potential



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1

FEASIBILITY

- **Accessible Technology:**
Web based secure platform for users, developed in React + Spring Boot, accessible through any modern browser.
- **Zero Barrier:**
WhatsApp bot enables you to report in local languages without installing any extra software.
- **Proven Infrastructure:**
PostGIS provides location-based analytics and Python AI provides the risk scoring.

2

VIABILITY

- **Government Adoption :**
High chance of adoption by State Animal Husbandry Departments and dairy cooperatives.
- **Economic Impact:**
Prevents and detects diseases early to reduce animal losses.
- **Long-Term Growth:**
Scalable platform with AI analytics, API integration and future growth.

3

CHALLENGES

- **Location Limitation:**
WhatsApp shared photographs do not have correct GPS data. Users must share the live location separately..
- **Scalable server:**
Serves many requests at the same time with predictable performance.
- **Low Connectivity:**
Ensuring reliable data sync and emergency alerts in low-network rural zones.

Outcomes & Strategic Benefits



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IMPACT

Disease Prevention :

Early detection helps prevent disease outbreaks.

Livestock Protection :

Reduces animal mortality and disease spread.

Economic Impact :

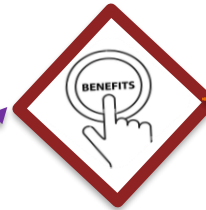
Minimizes financial losses for farmers and dairy sectors.

Public Health :

Supports faster response to zoonotic disease risks

Data-Driven Governance :

Enables authorities to monitor disease hotspots in real time.



BENEFITS

Easy Reporting :

Farmers can report through a familiar WhatsApp bot..

Live Location :

Accurate location helps identify affected areas.

AI Risk Scoring :

Provides quick disease-risk assessment.

Faster Response :

Helps connect cases with veterinary support quickly.

Centralized Dashboard :

Authorities can monitor cases, trends, and hotspots.

RESEARCH AND REFERENCES

- ☐ <https://pmc.ncbi.nlm.nih.gov/articles/PMC2865087>
- ☐ <https://www.mdpi.com/2624-7402/6/2/84>
- ☐ <https://www.sciencedirect.com/science/article/abs/pii/S1746809424013922>
- ☐ <https://www.annualreviews.org/content/journals/10.1146/annurev-animal-111523-102133>
- ☐ <https://www.sciencedirect.com/science/article/pii/S2772375526002017>
- ☐ <https://onlinelibrary.wiley.com/doi/full/10.1111/agec.70062>

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Biomedical Signal Processing and Control

Volume 102, April 2025, 107334

Livestock animal skin disease detection and classification using deep learning approaches

Dagne Walle Girmaw 

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<https://doi.org/10.1016/j.bspc.2024.107334>
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Highlights

- We proposed a Deep Learning Approaches for animal skin diseases

Open Access

Review

Advancing Livestock Technology: Intelligent Systemization for Enhanced Productivity, Welfare, and Sustainability

by Petru Alexandru Vlaicu ¹ , Mihail Alexandru Gras ^{2,*} , Arabela Elena Untea ¹ , Nicoleta Aurelia Lefter ³  and Mircea Catalin Rotar ² 

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ANNUAL REVIEW OF ANIMAL BIOSCIENCES

Volume 13, 2025

Review Article | Open Access

A One Health Approach to Reducing Livestock Disease Prevalence in Developing Countries: Advances, Challenges, and Prospects

[Jennifer K. Lane](#)¹, [Terra Kelly](#)², [Brian Bird](#)¹, [Erika Chenais](#)^{3,9}, [Annette Roug](#)^{4,5}, [Gema Vidal](#)³, [Rodrigo Gallardo](#)⁶, [Huaijun Zhou](#)⁷, [Grace VanHoy](#)⁸ and [Woutrina Smith](#)¹

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